**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 291 · HTML structures and forms

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does an HTML element supply to the browser?

Q2. Which part of a browser page contains displayed content?

Q3. What does HTTPS protect?

Q4. C# array int[] a={3,5,7}; append 9 conceptually using a new array and give sum.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

element — HTML start/end tags and content.

attribute — Property on HTML element.

form — Collects and submits user data.

label — Identifies a form control.

GET — HTTP method placing data in request URL.

## New learning

Understand — HTML uses nested elements to give structure and semantics: head contains metadata and body contains visible content.

Trace — Forms collect and submit inputs. The label's for value matches input's id; the input's name is the key submitted to the server.

Evaluate — With method GET, query parameters appear in the URL; don't use GET for confidential credentials. Client-side required validation improves usability but is not a substitute for server validation.

## Worked example

What is sent by a GET form whose input name is q and value is binary?

1. Predict first — The action sets the destination path '/find' and GET encodes form fields into URL query parameters.

2. Show method — The input name 'q' becomes the query key and 'binary' the value.

# We Do · Practise with guidance

Correct the accessible label connection in a form where for='email' but input id='user'.

```
<label for='email'>Email</label>
<input id='email' name='email' type='email'>
```

Predict first

The label for and input id must match exactly.

Show method

The email input type provides built-in browser constraints, but the server must validate again.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

State roles of html, head, body, form, input and label. [6]

### Question 2

Write a form that submits an age field by GET to /check and explain sent key. [6]

### Question 3

Explain two shortcomings of relying entirely on required and type='email' for security-sensitive form validation. [6]

# Exit Ticket · Answer before checking

Setting required on an HTML form is enough to secure a server from invalid data.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Document structure or semantics.

Q2. The HTML body.

Q3. Confidentiality and integrity of transport, with server authentication.

Q4. The new sequence is [3,5,7,9], sum 24; fixed-size arrays need allocation/copy or suitable list structure.

### You Do pages · Independent questions

Q1. Document root [1], metadata [1], rendered content [1], collection/submission [1], data field [1], accessible descriptive association [1].

Q2. form action='/check' method='get' [2]; label for age [1], input id age [1], name age [1], resulting query key age [1].

Q3. Client controls can be bypassed [2]; invalid/malicious requests can be sent directly [1]; validate on server [2], error handling/accessibility [1].

### Exit Ticket · Model answer

False. Server-side checks remain essential.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding